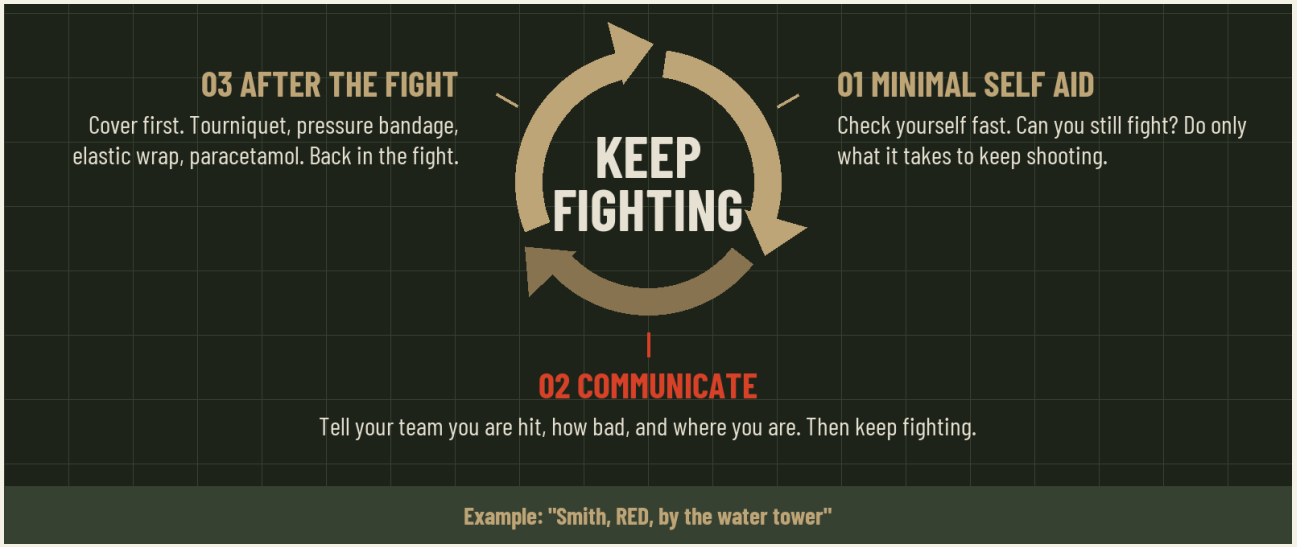
There is no first aid in a firefight. Win the fight, then provide aid.



SELF AID

1. Win the fight. Return fire, get to cover, then look at yourself.
2. Tourniquet any limb with many wounds.
3. Pressure bandage every other wound.
4. Elastic wrap once the bleeding has stopped.
5. Paracetamol for pain. 1 to 2 pills, works in about 4 minutes, lasts about 50.
6. Call CLS if you cannot stop the bleeding or a limb will not work.
7. Get back in the fight.

CALL CLS FOR

- Any unconscious casualty
- Any chest wound
- Bleeding you cannot stop
- A limb that will not work

BUDDY AID

1. Win the fight. Do not go to a casualty under fire.
2. Drag your buddy to cover. Treat behind cover.
3. Stop the bleeding. Head wounds first, then limbs, then the rest.
4. **Responsive?** Yes: finish bandaging, wrap, paracetamol, done. No: call CLS and keep going.
5. **Breathing?** No: turn the head to clear the mouth and check again.
6. **Pulse at the neck?** No: CPR. Yes: recovery position, stay with them.
7. Tell CLS what happened, how long they have been down, where the tourniquets are, pulse and breathing, and how many paracetamol.

YOU DO NOT

Airway adjuncts, chest seals, needles, IVs, any medication other than paracetamol.

1 BLEEDING

- **Limb with many wounds:** tourniquet. High and tight, above the wounds.
- **Every other wound:** pressure bandage. One per wound.
- **Bleeding stopped:** elastic wrap, so wounds do not reopen.
- Elastic wrap does not stop a bleeding wound on its own.

2 BREATHING

CHECK BREATHING	DO
Not breathing	Head turn, check pulse, call CLS
Shallow	Call CLS
Slow or rapid	Tell CLS
Normal	Recovery position

Blue skin on the face and arms: short of oxygen. Tell CLS.

3 BEATING

1. Check the pulse at the neck.
2. No pulse: CPR for 2 minutes.
3. Check the pulse again.
4. Repeat until the pulse returns or CLS takes over.

No pulse, no recovery position. It blocks CPR.

Weak or missing pulse at the arms or legs: heavy blood loss. Tell CLS.

ACM SMART CARD

JOINT MULTI-FUNCTIONAL STRIKE BATTALION

CLS

LEVEL 2 · MARCH

Treat in MARCH order. Do not move to the next letter until the current one is controlled.

M MASSIVE HEMORRHAGE

- **Limb, many wounds:** tourniquet.
- **Body part, many wounds:** emergency trauma dressing. Not on yourself.
- **Single wound:** pressure bandage.
- **Once stable:** elastic wrap over bandaged and clotted wounds.
- **Bruising, or bleeding that will not stop:** call the Medic for TXA.
- Tourniquet off once the wounds under it are bandaged.

A AIRWAY

PROBLEM	DO
Tongue blocks airway	NPA or OPA, head-tilt chin-lift, or recovery position
Vomit or blood	Head turn if mild. Suction bag
Collapse, long unconscious	Head-tilt chin-lift or recovery position. Medic for i-gel
Chemical inflammation or spasm	Medic. Needs a surgical airway

Suction first. NPA and OPA only go in a clear airway.

R RESPIRATION

INSPECT CHEST	MEANS	DO
Sides uneven	Pneumothorax	Chest seal
Tracheal deviation	Tension pneumothorax	Chest seal, then NCD
Bruising on upper chest	Hemothorax	Chest seal, call Medic
No movement	Not breathing	Check airway and pulse

PULSE OXIMETER

UNDER 80	UNDER 67	UNDER 55
Unconscious	Cardiac arrest	Death

With systolic under 90 the reading can be off by up to 30%.

C CIRCULATION

CHECK	RESULT	DO
Carotid pulse	None	Cardiac arrest drill
Radial or femoral	Weak or none	Medic for fluids
Body parts	Bruising	Medic for TXA

CARDIAC ARREST DRILL

1. Fix what you can: tension pneumothorax (seal and decompress), blocked airway (clear it).
2. Morphine or fentanyl on board: naloxone now.
3. Connect the AED, select AED mode.
4. CPR for 2 minutes or until the AED prompts. Let it re-analyze. Repeat.
5. No AED: CPR for 2 minutes, carotid pulse check, repeat.

H HEAD AND HYPOTHERMIA

- **Unconscious, stable vitals:** ammonia inhalant.
- **Still unconscious:** go back through M, A, R, C.
- **Conscious:** pain, then fractures.

FRACTURES

SAM splint, then elastic wrap over it. Significant swelling is a complex fracture: evacuate. Realignment is the Medic's.

NOT CLS

IV or IO, fluids, blood, vial medications, i-gel, surgical airway, thoracostomy, manual AED, fracture realignment, casualty conversion.

MEDS

MED	FOR	DOSE	ONSET	LASTS	WATCH
Paracetamol	Mild to moderate pain	1 to 2 pills	Under 4 min	About 50 min	
Penthrox inhaler	Moderate to severe pain, short term	1 to 5 uses, 8 per inhaler	Under 3 s	About 4 min	Not with chest injury, slow breathing, low blood volume or altered mental status
Naloxone spray	Opioid overdose	1 to 2 uses	Under 3 s	About 6 min	Opioids last longer. Watch and re-dose
Ammonia inhalant	Unconscious with stable vitals	1 use, 8 per inhalant	Under 3 s	Under 40 s	
ATNA autoinjector	Nerve agent	2 to 3 injections to start	Under 15 s	About 15 min	
Midazolam autoinjector	Seizures from nerve agent	1 injection	Under 30 s	About 20 min	Lowers heart rate and blood pressure

Everything CLS does, plus anything that needs a vein, a scalpel or a rhythm decision.

M MASSIVE HEMORRHAGE

- **TXA 1 g IV** for severe bleeding, noticeable bruising or a bleeding hemothorax.
- Bruising with a falling blood pressure is internal bleeding. TXA and fluids.
- Clots are unstable. Wrap or stitch once stable.

A AIRWAY

TOOL	SOLVES
i-gel	Open airway, collapse. Airway must be clear first
ACCUVAC, suction bag	All obstructions
Surgical airway (CricKit)	Collapse, obstruction, inflammation, spasm

Use the least invasive airway that solves the problem.

R RESPIRATION

- **Stethoscope:** muffled on one side is a pneumothorax.
- **Tension pneumothorax:** chest seal plus NCD, or thoracostomy.
- **Hemothorax:** TXA, then thoracostomy and chest tube.
- Consider lidocaine IM before thoracostomy.

FINGER THORACOSTOMY	DO
Lung inflating normally	Close incision
Lung not inflating or collapsed	Close incision
Blood in pleural space	Chest tube
Active bleeding	TXA, then chest tube

Chest tube: drain with the ACCUVAC or a suction bag, re-drain until no blood comes out, close the incision.

H HEAD, PAIN, FRACTURES

- **Unconscious, stable vitals:** ammonia inhalant.
- **Still unconscious:** go back through M, A, R, C.
- **Fracture:** lidocaine 50 to 100 mg IM, realign, SAM splint, check mobility, wrap.
- **Complex fracture:** evacuate.

C ACCESS

ACCESS	SITE	FLOW
16g IV	Limbs	Moderate
14g IV	Limbs	Fast. More complications
FAST1 IO	Sternum	Slow
EZ-IO	Tibia or humerus	Slow

Crashing casualty: IO first, then move the bags to an IV. Tourniquets block IVs on that limb.

C FLUIDS

FLUID, BEST FIRST	CARRIES OXYGEN	FLOW	PLATELETS
Fresh whole blood	Yes	Fast	Yes, with calcium
Whole blood	Yes	Slow	Yes, with calcium
Plasma	30%	Moderate	Yes
Saline	No	Fast	No

BLOOD

- Check the blood type on the dog tag. A mismatch can kill.
- O- goes to anyone. AB+ receives from anyone. A negative recipient only takes negative blood.
- **Calcium chloride 1 g IV** after the first unit (500 ml), then after every 4 units. Never IM.

FIELD TRANSFUSION

1. Connect the kit to the donor's IV. It does not work on an IO.
2. Wait for it to fill.
3. Disconnect and use it as fresh whole blood, soon.

CASUALTY CONVERSION

- Casualty unconscious, alive and not bleeding externally.
- If the mission requires it: in a medical facility or vehicle, and ertapenem 1 g given.
- Keep treating the AI casualty until it is evacuated.

C CARDIAC ARREST

Confirm with a carotid pulse check. Fix the reversible cause. Then rhythm, drug, shock, CPR.

REVERSIBLE CAUSE	DO
Hypovolemia	IO, push fluid, CPR
Hypoxia	CPR with BVM
Tension pneumothorax	NCD or thoracostomy
Hemothorax	TXA, thoracostomy, chest tube
Opioids on board	Naloxone

RHYTHM	SHOCK	DO
Sinus	No	Re-assess
VT or VF	Yes	Amiodarone or lidocaine, shock, CPR
PEA	No	Re-check reversible causes, CPR
Asystole	No	Check for terminal injuries, epinephrine, CPR

PEA looks like sinus on the EKG. Always confirm with a pulse check.

ARREST DRUGS

- **Epinephrine 1 mg IV.** The only drug that helps in asystole.
- **Amiodarone 300 mg IV** for VT or VF. Two vials. Less does nothing.
- **Lidocaine 100 mg IV** for VT or VF. One vial.
- Never amiodarone and lidocaine together. Leave 5 to 10 minutes between them, and between amiodarone doses.

CPR

- 2 minutes, or until the AED prompts. Re-analyze. Repeat.
- **Short of oxygen, one medic or no i-gel:** 30 compressions, then 2 BVM breaths.
- **Short of oxygen, two medics and an i-gel:** continuous CPR and BVM.
- **Pulse is back:** EtCO2 jumps to 40 to 50, cyanosis fades, carotid pulse returns.

Draw = dose ÷ concentration. IV drugs need an IV or IO first. More than the range adds side effects, not benefit.

FIXED DOSES

DRUG	FOR	DOSE	DRAW
Epinephrine	Cardiac arrest	1 mg IV	1 ml
Amiodarone	VT or VF	300 mg IV	6 ml, two vials
Lidocaine	VT or VF	100 mg IV	5 ml
Lidocaine	Before realignment or thoracostomy	50 to 100 mg IM	2.5 to 5 ml
TXA	Severe bleeding, bruising, bleeding hemothorax	1 g IV, up to 2 g	10 ml
Calcium chloride	After the first unit of blood, then every 4 units	1 g IV, never IM	10 ml
Ondansetron	Nausea, vomiting, after ketamine	4 mg IV or IM, max 8 mg	2 ml
Ertapenem	Wound infection, before conversion if required	1 g IV or IM	3.2 ml
Atropine	Nerve agent	2 mg IV or IM, 2 to 4 mg to start	2 ml, two vials
Atropine	Slow heart rate after amiodarone	1 mg IV or IM	1 ml
Adenosine	Lowers heart rate	6 mg IV	2 ml

WEIGHT DOSES, WORKED FOR 80 KG

DRUG	ROUTE	PER KG	AT 80 KG	DRAW
Morphine	IV	0.05 to 0.1 mg	4 to 8 mg	0.8 to 1.6 ml
Morphine	IM	0.07 to 0.1 mg	5.6 to 8 mg	1.1 to 1.6 ml
Fentanyl	IV	0.5 to 1 mcg	40 to 80 mcg	0.8 to 1.6 ml
Fentanyl	IM	0.7 to 1 mcg	56 to 80 mcg	1.1 to 1.6 ml
Ketamine	IV	0.1 to 0.2 mg	8 to 16 mg	0.16 to 0.32 ml
Ketamine	IM	0.4 to 0.8 mg	32 to 64 mg	0.64 to 1.28 ml
Lidocaine	IV	1 mg, max 3 mg	80 mg, max 240 mg	4 ml, max 12 ml
Esmolol	IV	0.25 to 0.5 mg	20 to 40 mg	2 to 4 ml

TIMINGS AND VITALS

DRUG	ONSET IV / IM	LASTS	VITALS
Morphine	Under 25 s / 2 min	21 / 30 min	Lowers HR, BP, breathing
Fentanyl	Under 10 s / 30 s	16 / 20 min	Lowers HR, BP, breathing
Ketamine	Under 5 s / 20 s	11 / 15 min	None
Fentanyl lozenge	Under 2 min	15 min	Lowers breathing most
Epinephrine	Under 5 s / 30 s	6 to 7 min	Raises HR, BP, breathing
Amiodarone	Under 10 s	12 min	Lowers HR, BP
Lidocaine	Under 5 s	10 min	Lowers HR, BP
Atropine	Under 5 s / 15 s	15 min	Raises HR
Esmolol	Under 20 s	5 min	Lowers HR
Adenosine	Under 1 min	Under 2 min	Lowers HR
TXA	Under 15 s	15 min	Lowers BP slightly
Calcium chloride	Under 15 s	10 min	
Ondansetron	Under 15 s / 45 s	12 to 15 min	
Ertapenem	Under 5 s / 30 s	15 to 20 min	

DO NOT MIX

- No amiodarone and lidocaine within 5 to 10 minutes of each other. No second amiodarone within 5 to 10 minutes.
- No opioids with low blood volume, slow heart rate, slow breathing, chest injury or altered mental status.
- Naloxone for any arrest with opioids on board.
- Calcium chloride never IM, and not in ventricular fibrillation.
- No epinephrine with a very fast heart rate or high blood pressure.
- No esmolol, adenosine, amiodarone or midazolam with a slow heart rate.
- Morphine autoinjector is not issued.

PAIN: WHICH DRUG

SITUATION	DRUG
Breathing impaired or chest injury	Ketamine. Consider ondansetron 4 mg after
No breathing problems	Morphine. Fentanyl if severe
Conscious, lying down, no IV	Fentanyl lozenge, 1